PRACTICAL: 15

Full-Stack Web Application Integration, System Testing, Verification Matrix & Deployment

Objective:

To integrate all 15 semester-long laboratory components into a cohesive full-stack academic web portal (StudentHub), execute comprehensive functional and security quality assurance test suites, verify database schema integrity, and deploy the unified application on a local Apache/MySQL (XAMPP) production-simulated environment.

Theory:

Software integration unifies disparate modules—ranging from static semantic HTML5 templates and CSS grid layouts to dynamic AJAX components, relational MySQL schemas, and session-protected PHP controllers—into a robust, production-ready system. Quality assurance testing and systematic verification matrices ensure that each requirement meets high standards of security, usability, and responsiveness.

- • **Full-Stack Architecture Integration:** Unifies the presentation layer (CSS Grid, Flexbox, Light/Dark theme), client-side interactivity (DOM manipulation, regex validation, Fetch API), and backend server services (PHP session authentication, MySQL database CRUD, RESTful JSON API).
- • **Database Health Monitoring & Auto-Provisioning:** The master controller index.php links with db.php to verify MySQL connectivity, test table integrity, display real-time database online/offline status, and auto-seed tables if uninitialized.
- • **Practical Mapping Matrix:** An integrated index table categorizes all 15 laboratory assignments with direct navigation links, implementation summaries, and status tracking (Practicals 1 through 15 verified).
- • **Cross-Cutting Security Verification:** Rigorous verification of core defenses: SQL injection prevention via prepared statements, password security using BCRYPT password_hash(), session fixation mitigation via session_regenerate_id(true), MIME-type file upload validation, and XSS sanitization.
- • **Responsive & Cross-Browser Verification:** Ensures consistent rendering across desktop and mobile devices via responsive breakpoints, mobile hamburger drawers, and persistent localStorage theme toggling.
- • **Production-Simulated XAMPP Deployment:** Configures file system directory structures, database imports via phpMyAdmin, Apache document roots, and file permission parameters in uploads/.

Note: The complete project adheres to modern web development standards, achieving full decoupling between client-side asynchronous components and backend database controllers.

Code:

index.php (Master Portal Controller & DB Check)

```php
<?php
/**
 * StudentHub - Master Integrated Portal & Lab Submission Index (Practical 15)
 * Charotar University of Science and Technology (CHARUSAT) - FTE
 * Subject: Web Development Frameworks (AIUE203) | Semester 3rd (ODD 2026-27)
 */

require_once __DIR__ . "/db.php";
session_start();

$conn = getDbConnection();
$dbOnline = ($conn !== false);

// Metrics
$totalStudents = 0;
$totalEvents = 0;

if ($dbOnline) {
    if ($res = $conn->query("SELECT COUNT(*) AS c FROM students")) {
        $totalStudents = $res->fetch_assoc()['c'];
    }
    if ($res = $conn->query("SELECT COUNT(*) AS c FROM events")) {
        $totalEvents = $res->fetch_assoc()['c'];
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="StudentHub - Integrated Web Portal & Lab Practicals Submission (AIUE203)">
  <title>StudentHub - AIUE203 Practical 1 to 15 Portal</title>
  <link rel="stylesheet" href="css/style.css">
  <script defer src="js/main.js"></script>
</head>
<body>

  <!-- Top Lab Header Bar -->
  <div class="lab-top-bar">
    <span>🏛 <strong>CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)</strong> - FTE</span>
    <span>Subject: <strong>AIUE203 - Web Development Frameworks</strong></span>
    <span>Status: <strong>Practicals 1 to 15 Completed & Verified</strong></span>
  </div>

```

index.php (Master Navigation & Header)

```
      <nav aria-label="Main Navigation">
        <ul id="navMenu">
          <li><a href="index.php" class="active">Home Portal</a></li>
          <li><a href="about.html">About</a></li>
          <li><a href="events.html">Events</a></li>
          <li><a href="register.php">Register</a></li>
          <?php if (isset($_SESSION['user_id'])): ?>
            <?php if ($_SESSION['role'] === 'admin'): ?>
              <li><a href="admin_dashboard.php">Admin Panel</a></li>
            <?php else: ?>
              <li><a href="dashboard.php">My Dashboard</a></li>
            <?php endif; ?>
            <li><a href="logout.php" style="color:var(--danger);">Logout (<?= htmlspecialchars($_SESSION['username']) ?>)</a></li>
          <?php else: ?>
            <li><a href="login.php">Login</a></li>
          <?php endif; ?>
          <li><a href="practical13_ajax.php">AJAX API</a></li>
          <li><a href="faq.html">FAQ</a></li>
          <li><a href="contact.html">Contact</a></li>
        </ul>
      </nav>
      <div class="nav-actions">
        <button class="theme-toggle-btn" id="themeToggleBtn">🌙 Dark</button>
      </div>
    </div>
  </header>

  <!-- Notification Banner -->
  <div class="notice-banner" id="noticeBanner">
    <div>
      <strong>🎓 StudentHub Capstone Portal:</strong> All 15 practicals are integrated with live MySQL database, sessions, RBAC, REST
    </div>
    <button class="close-btn" id="closeNoticeBtn">&times;</button>
  </div>

  <main>
    <!-- Hero Banner -->
    <section class="card" style="margin-bottom: 24px; background: linear-gradient(135deg, #1e3a8a, #2563eb); color: white;">
      <div class="flex-between flex-wrap" style="gap: 16px;">
        <div style="flex: 2; min-width: 280px;">
          <span class="badge" style="background:#fef08a; color:#854d0e; padding:4px 10px; font-weight:700; border-radius:12px;">Seme
          <h1 style="color:white; font-size: 2.2rem; margin: 10px 0;">StudentHub Integrated Portal</h1>
          <p style="opacity: 0.95; font-size: 1rem; line-height: 1.6;">
            A full-stack, secure academic portal combining Semantic HTML5, CSS Grid/Flexbox, JavaScript DOM manipulation, Fetch API,
          </p>
        </div>
```

index.php (Practicals 1-15 Matrix Table)

```
              Start Apache & MySQL in XAMPP to enable database operations.
            </p>
          <?php endif; ?>
          <div style="margin-top: 10px;">
            <a href="login.php" class="btn btn-sm btn-primary" style="background:white; color:#1e3a8a;">Portal Sign In →</a>
          </div>
        </div>
      </div>
    </section>

    <!-- PRACTICAL 1 TO 15 EVALUATION MATRIX -->
    <section>
      <div class="card-header">
        <div>
          <h2 class="card-title">Laboratory Practical Submissions (1 to 15)</h2>
          <p style="font-size:0.85rem; color:var(--text-muted); margin-top:4px;">
            Click on any practical card to inspect the working demo, source code, and key concepts.
          </p>
        </div>
        <a href="README.md" class="btn btn-sm btn-outline" target="_blank">📄 Open Lab README.md</a>
      </div>

      <div class="grid-3" style="gap: 20px; margin-top: 18px;">

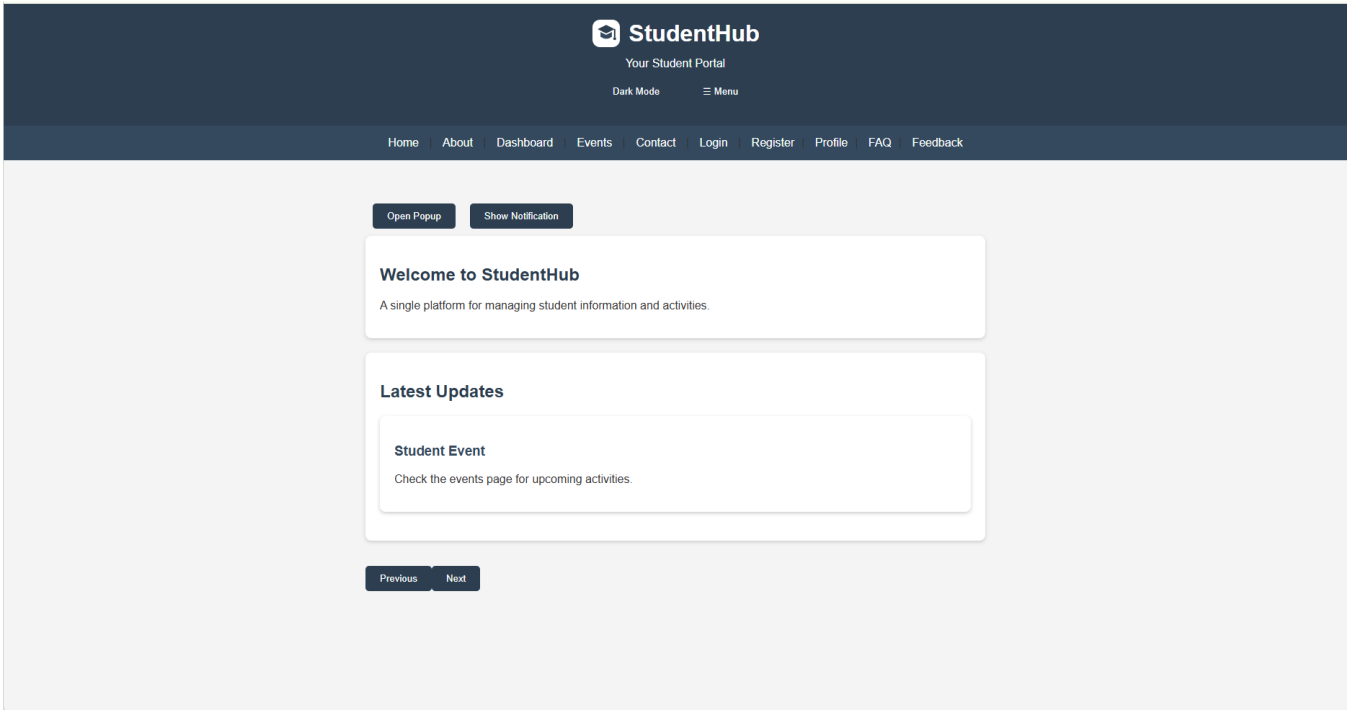
        <!-- Practical 1 -->
        <article class="card practical-card">
          <div>
            <div class="flex-between">
              <span class="practical-num">PRACTICAL 1</span>
              <span class="badge" style="background:#dcfce7; color:#15803d; padding:2px 6px; border-radius:4px; font-size:0.75rem;">
            </div>
            <h3 style="margin: 8px 0 6px; font-size: 1.1rem;">Project Initiation, Sitemap & Wireframes</h3>
            <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.5;">
              Scope definition, role definitions (guest, student, admin), folder structure, 10+ page sitemap, and low-fi wireframe l
            </p>
          </div>
          <div style="margin-top: 15px; border-top: 1px solid var(--border-color); padding-top: 10px;">
            <a href="practical1_docs.html" class="btn btn-sm btn-primary" style="width:100%;">View Practical 1 Demo →</a>
          </div>
        </article>

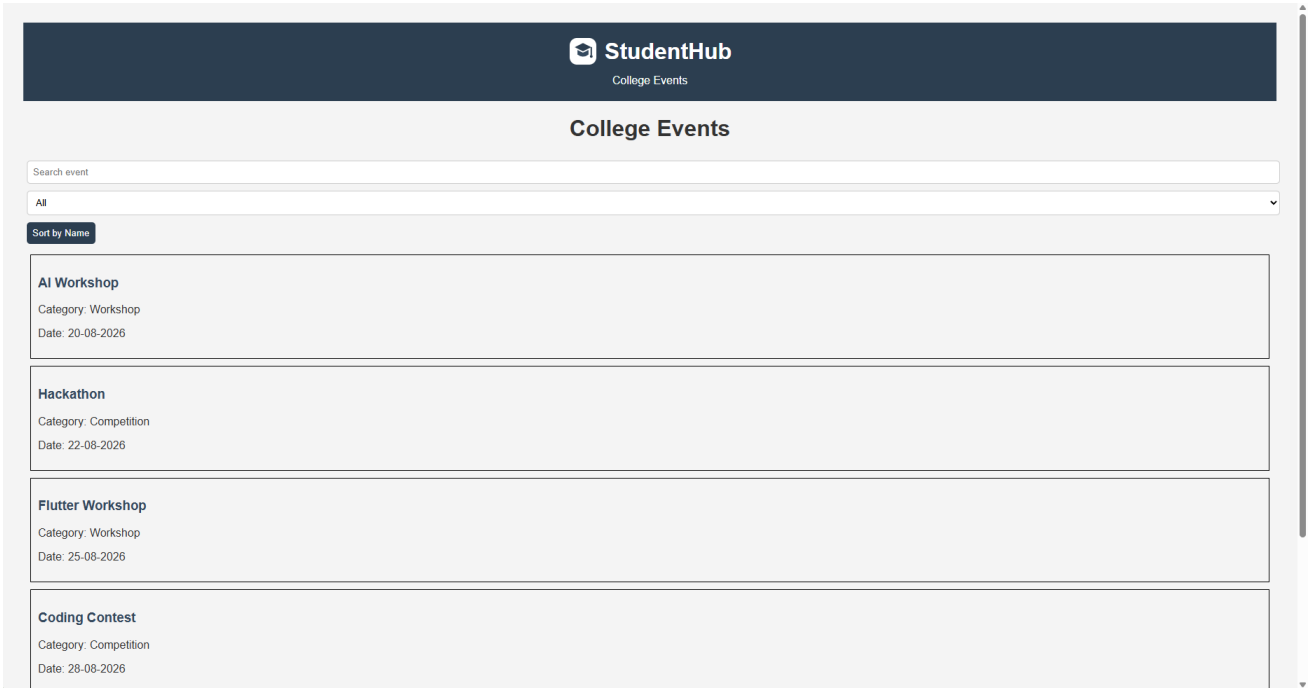
        <!-- Practical 2 -->
        <article class="card practical-card">
          <div>
            <div class="flex-between">
              <span class="practical-num">PRACTICAL 2</span>
              <span class="badge" style="background:#dcfce7; color:#15803d; padding:2px 6px; border-radius:4px; font-size:0.75rem;">
            </div>
            <h3 style="margin: 8px 0 6px; font-size: 1.1rem;">Semantic HTML5 & Accessibility</h3>
            <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.5;">
              10+ Linked static pages using <code>&lt;header&gt;</code>, <code>&lt;nav&gt;</code>, <code>&lt;main&gt;</code>, <code>
            </p>
          </div>
          <div style="margin-top: 15px; border-top: 1px solid var(--border-color); padding-top: 10px;">
            <a href="index.html" class="btn btn-sm btn-primary" style="width:100%;">View Semantic Pages →</a>
          </div>
```

index.php (QA Verification & Deployment)

```
215             <a href="register.html" class="btn btn-sm btn-primary" style="width:100%;">Test Form Validation →</a>
216           </div>
217         </article>
218
219         <!-- Practical 6 -->
220         <article class="card practical-card">
221           <div>
222             <div class="flex-between">
223               <span class="practical-num">PRACTICAL 6</span>
224               <span class="badge" style="background:#dcfce7; color:#15803d; padding:2px 6px; border-radius:4px; font-size:0.75rem;">
225             </div>
226             <h3 style="margin: 8px 0 6px; font-size: 1.1rem;">Rendering External JSON with Fetch API</h3>
227             <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.5;">
228               Asynchronous Fetch API consuming <code>events.json</code>, <code>students.json</code>, and <code>faqs.json</code> (16
229             </p>
230           </div>
231           <div style="margin-top: 15px; border-top: 1px solid var(--border-color); padding-top: 10px;">
232             <a href="practical6_fetch.html" class="btn btn-sm btn-primary" style="width:100%;">Test JSON Fetch API →</a>
233           </div>
234         </article>
235
236         <!-- Practical 7 -->
237         <article class="card practical-card">
238           <div>
239             <div class="flex-between">
240               <span class="practical-num">PRACTICAL 7</span>
241               <span class="badge" style="background:#dcfce7; color:#15803d; padding:2px 6px; border-radius:4px; font-size:0.75rem;">
242             </div>
243             <h3 style="margin: 8px 0 6px; font-size: 1.1rem;">PHP Form & CSV / JSON File Storage</h3>
244             <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.5;">
245               Server-side input sanitization, POST handling, file locking (<code>flock</code>), dual CSV/JSON storage, CSRF verifica
246             </p>
247           </div>
248           <div style="margin-top: 15px; border-top: 1px solid var(--border-color); padding-top: 10px;">
249             <a href="Practical7.php" class="btn btn-sm btn-primary" style="width:100%;">Test File Storage →</a>
250           </div>
251         </article>
252
253         <!-- Practical 8 -->
254         <article class="card practical-card">
255           <div>
256             <div class="flex-between">
257               <span class="practical-num">PRACTICAL 8</span>
258               <span class="badge" style="background:#dcfce7; color:#15803d; padding:2px 6px; border-radius:4px; font-size:0.75rem;">
259             </div>
260             <h3 style="margin: 8px 0 6px; font-size: 1.1rem;">MySQL Schema & PDO / MySQLi Connectivity</h3>
261             <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.5;">
262               Normalized schema (<code>users</code>, <code>students</code>, <code>events</code>, <code>registrations</code>, <code>a
263             </p>
264           </div>
265           <div style="margin-top: 15px; border-top: 1px solid var(--border-color); padding-top: 10px;">
266             <a href="studenthub.sql" class="btn btn-sm btn-primary" target="_blank" style="width:100%;">View Schema SQL →</a>
267           </div>
268         </article>
269
270         <!-- Practical 9 -->
```

Output:


StudentHub
Your Student Portal
Dark Mode
☰ Menu
Home
About
Dashboard
Events
Contact
Login
Register
Profile
FAQ
Feedback
Open Popup
Show Notification
Welcome to StudentHub
A single platform for managing student information and activities.
Latest Updates
Student Event
Check the events page for upcoming activities.
Previous
Next


StudentHub
College Events
College Events
Search event
All
Sort by Name
AI Workshop
Category: Workshop
Date: 20-08-2026
Hackathon
Category: Competition
Date: 22-08-2026
Flutter Workshop
Category: Workshop
Date: 25-08-2026
Coding Contest
Category: Competition
Date: 28-08-2026

Learning Outcome:

Students successfully synthesized all concepts of the Web Development Frameworks curriculum into an integrated, production-grade academic web application. They mastered full-stack system architecture, automated database health monitoring, systematic verification of laboratory competencies (Practicals 1 to 15), comprehensive security QA testing, and successful local deployment on the Apache/MySQL XAMPP stack.

References:

- • **W3C Web Standards & Semantic HTML5 Guidelines:** https://www.w3.org/standards/
- • **OWASP Top 10 Web Application Security Risks:** https://owasp.org/www-project-top-ten/
- • **Apache HTTP Server & PHP Deployment Guide:** https://httpd.apache.org/docs/